

1.9 Classwork: Bivariate data and linear regression — Markscheme

Calculator allowed. Marks are not printed on the student sheet; they are recorded here. Total: 18 marks.

GDC: enter x in one list and y in a second, then **menu** $\rightarrow$ **Statistics** $\rightarrow$ **Stat Calculations** $\rightarrow$ **Linear Regression** (**$mx+b$**). The screen gives the gradient as m (the a of $y = ax + b$), the intercept b , r^2 and r . For the line of x on y , run it again with the **X List** and **Y List** swapped.

1. [Maximum mark: 9]

$$n = 8, \quad \bar{x} = 24.625, \quad \bar{y} = 204.25.$$

(a) $r = 0.98212 \dots = 0.982$ (3 s.f.) **A1**

Do not accept $r^2 = 0.965$, which sits beside r on the regression screen.

(b) Strong, positive (linear) correlation **A1A1**

A1 for “strong” (accept “very strong”), A1 for “positive”; the two marks are independent. “Moderate” or “weak” earns A0 for the strength: $|r| = 0.982$ is well above 0.8. FT from the candidate’s r in (a).

(c) $y = 8.37x - 1.96$ **A1A1**

A1 for $a = 8.37$, A1 for $b = -1.96$; the two marks are independent. Accept $a = 8.3738 \dots$, $b = -1.9559 \dots$, or either value to more figures. Award A1A0 for $8.37x + 1.96$ (sign of b lost). The reversed line $x = 0.115y + 1.10$ earns A0A0.

(d) For each increase of 1°C in the maximum temperature, the gelateria sells about 8.37 more gelati. **A1**

The answer must give the rate (about 8 or 8.37 gelati) per degree, in context. “As the temperature rises, more gelati are sold” restates the correlation without the rate and earns A0. FT from the candidate’s a in (c).

(e) $y = 8.37(26) - 1.96$ **(M1)**

$= 215.76 \dots \approx 216$ gelati **A1**

*Accept any value consistent with the candidate’s own correctly-rounded coefficients, including 215.76... (full precision) and 215.66 (from 8.37 and -1.96), and either rounded to 216. FT from the candidate’s line in (c). **Feedback:** store the coefficients in the GDC (or use the regression function **f1(26)**) and round only the final answer.*

(f) 10°C is outside the range of the recorded temperatures (18 to 32°C), so the estimate would be an extrapolation and is not reliable. **R1**

R1 for stating that 10 lies outside the data range (or “below the lowest temperature recorded”). Accept “extrapolation” with a reference to the data. “The correlation might not hold in winter” with no reference to the range of the data earns R0. Computing $y = 81.8$ is not required and earns nothing on its own.

2. [Maximum mark: 9]

$$n = 8, \quad r = -0.803 \text{ (3 s.f.)}.$$

(a) $y = -0.344x + 8.88$

A1A1

A1 for $a = -0.344$, A1 for $b = 8.88$; the two marks are independent. Accept $a = -0.34370\dots$, $b = 8.8834\dots$, or either value to more figures. Award A0 for the gradient if its negative sign is lost.

(b) A student who spends no time on their phone is predicted to sleep about 8.88 hours a night.

A1

The answer must refer to zero phone time (or “ $x = 0$ ”) and to sleep in hours. “The y -intercept is 8.88” with no context earns A0. Accept a comment that $x = 0$ lies outside the data, so the value is itself an extrapolation, alongside the interpretation. FT from the candidate’s b in (a).

(c) $(\bar{x}, \bar{y}) = (3.625, 7.6375)$

A1A1

*A1 for $\bar{x} = 3.625$, A1 for $\bar{y} = 7.6375$; the two marks are independent. Accept 3.63 and 7.64 (3 s.f.). **GDC:** One-Variable Statistics on each list, or Two-Variable Statistics on both. By hand, $\bar{x} = \frac{29}{8}$ and $\bar{y} = \frac{61.1}{8}$. Both regression lines pass through the mean point.*

(d) $x = -1.88y + 18.0$

A1A1

A1 for the gradient -1.88 , A1 for the intercept 18.0; the two marks are independent. Accept $-1.8756\dots$ and $17.950\dots$ (17.95). Rearranging the line from (a) to make x the subject, $x = -2.91y + 25.8$, is not the regression line of x on y and earns A0A0.

(e) $x = -1.88(7.0) + 18.0$

M1

$$= 4.8207\dots = 4.82 \text{ hours (3 s.f.)}$$

A1

*M1 for substituting $y = 7.0$ into the line of x on y : the estimate is of x from a given y , so the line of x on y is the appropriate one. Substituting into the line of y on x and solving, $7.0 = -0.344x + 8.88$, giving $x = 5.48$, is the wrong line and earns M0A0. Accept any value consistent with the candidate’s own correctly-rounded coefficients, including 4.8207\dots (full precision), 4.79 (from -1.88 and 17.95) and 4.84 (from -1.88 and 18.0). FT from the candidate’s line in (d). **Feedback:** where coefficients were rounded before substituting, store them in the GDC and round only the final answer.*